

SCHOOL OF COMPUTER SCIENCE & ENGINEERING

Department of Computer Science & Application

WEEKLY PROGRESS REPORT (EXTENDED TECHNICAL REPORT)

Class & Semester: B.Tech CSE / Final Year (Sem VII/VIII)	Week No.: Week 2 Date: Aug 11 - Aug 16, 2025
Project Title: Rankly.ai – AI-Powered Automated Recruitment Intelligence & ATS Platform	
Module Focus: Week 2: Database Modeling, Schema Architecture & Backend Foundation	

Detailed Work Completed / Technical Implementation:

- Relational Database Architecture: Engineered a scalable relational data model using Prisma ORM with PostgreSQL (and SQLite development fallback). Defined comprehensive entities: User (with Role enums: CANDIDATE, RECRUITER, ADMIN), Organization, JobRequisition, CandidateProfile, ResumeDocument, SkillAssessment, OtpVerificationToken, and AuditLog.
- Relational Integrity & Indexing: Established Foreign Key cascades, unique compound indexes on (orgId, jobCode) and (userId, email), and B-Tree indexes on search query attributes (skills, location, status) to ensure sub-10ms query execution times.
- Express REST API Gateway: Built a modular Node.js/Express server architecture featuring structured routing (/api/auth, /api/candidates, /api/jobs, /api/analytics, /api/health) with request validation pipelines.
- Authentication & Cryptographic Security: Configured JSON Web Token (JWT) asymmetric signing with configurable TTL, bcrypt salt-round 10 password hashing, Helmet security headers, CORS origin whitelisting, and centralized error handling middleware.
- Database Migration Pipeline: Created automated Prisma migration scripts (prisma migrate dev), database seed scripts with mock enterprise data, and connection pooling configurations for Railway deployment.

Key Deliverables & Verified Output: Prisma schema with 8 relational models, 14 foreign keys, complete REST API gateway with 16 core endpoints, and verified database seed scripts.

Guide's Remark: Excellent technical depth. The architecture diagrams, code implementations, and security standards demonstrate thorough milestone completion.

PRN No.	Name of Present Student	Signature
2103001	Mayur Patil	

Project Guide

Project Coordinator

Associate Dean, CSA

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WEEKLY PROGRESS REPORT (EXTENDED TECHNICAL REPORT)

Class & Semester: B.Tech CSE / Final Year (Sem VII/VIII)	Week No.: Week 3 Date: Aug 18 - Aug 23, 2025
Project Title: Rankly.ai – AI-Powered Automated Recruitment Intelligence & ATS Platform	
Module Focus: Week 3: Dual-Persona Authentication, Onboarding & Multi-Tenant Security	

Detailed Work Completed / Technical Implementation:

- Dual-Persona Sliding Auth Architecture: Designed and built an interactive sliding authentication UI featuring two distinct user workflows (Candidate Registration vs Enterprise Recruiter Workspace Sign-In) inside a single fluid container.
- 3D Card Flip & Magic UI Integration: Integrated full 3D card flipping CSS transforms (perspective: 1000px, transform-style: preserve-3d) alongside Magic UI animated luminous border beams (--border-beam-from: #D95D39, --border-beam-to: #5B998C) with zero UI layout shift.
- Email OTP Verification Engine: Implemented a secure OTP generation service generating cryptographically random 6-digit verification codes, stored with SHA-256 hashes and 10-minute expiry timestamps in PostgreSQL.
- Anti-Abuse & Rate-Limiting Controls: Engineered sliding-window rate limiters restricting OTP requests to maximum 3 attempts per 15 minutes per IP/email, complete with real-time UI countdown timers and resend throttling.
- Role-Based Access Control (RBAC): Developed Express authorization middleware (verifyRole(['RECRUITER']), verifyRole(['CANDIDATE'])) strictly preventing unauthorized privilege escalation across tenant boundaries.

Key Deliverables & Verified Output: Production-ready sliding auth card with 3D flip physics, end-to-end Email OTP verification pipeline with rate-limiting, and strict RBAC authorization middleware.

Guide's Remark: Excellent technical depth. The architecture diagrams, code implementations, and security standards demonstrate thorough milestone completion.		
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WEEKLY PROGRESS REPORT (EXTENDED TECHNICAL REPORT)

Class & Semester: B.Tech CSE / Final Year (Sem VII/VIII)	Week No.: Week 4 Date: Aug 25 - Aug 30, 2025
Project Title: Rankly.ai – AI-Powered Automated Recruitment Intelligence & ATS Platform	
Module Focus: Week 4: Logo.dev Brand Lookup Integration & Enterprise Recruiter Workspace	

Detailed Work Completed / Technical Implementation:

- Logo.dev API Architecture: Integrated Logo.dev API (pk_EJiYI6zfSfSYg_x25fXJSg) via a custom client-side caching engine and debounce handler. Implemented dual lookup strategy: domain-based (img.logo.dev/{domain}) and company-name-based (img.logo.dev/name/{companyName}) with high-DPI Retina support (&retina=true) and SVG/PNG fallbacks.
- Enterprise Recruiter Dashboard: Built a multi-section employer portal comprising Organization Profile Setup, Department Hierarchy, Hiring Stage Configuration, and Team Member Access permissions.
- Multi-ATS Connector Architecture: Designed integration presets and data normalization schema for major ATS vendors (Greenhouse, Workday, Lever, and Ashby), enabling bi-directional job requisition import and candidate stage sync.
- Job Requisition Posting Engine: Engineered dynamic job creation forms with custom screening criteria, mandatory vs preferred skill tagging, salary benchmarking parameters, and calibrated evaluation weighting sliders.

Key Deliverables & Verified Output: Real-time Logo.dev brand auto-sync integrated across employer onboarding, Recruiter Workspace UI with multi-department job manager, and ATS connector configuration matrix.

Guide's Remark: Excellent technical depth. The architecture diagrams, code implementations, and security standards demonstrate thorough milestone completion.		
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WEEKLY PROGRESS REPORT (EXTENDED TECHNICAL REPORT)

Class & Semester: B.Tech CSE / Final Year (Sem VII/VIII)	Week No.: Week 5 Date: Sep 01 - Sep 06, 2025
Project Title: Rankly.ai – AI-Powered Automated Recruitment Intelligence & ATS Platform	
Module Focus: Week 5: AI Resume Parser, Gemini LLM Semantic Matcher & Automated Skill Gap Matrix	

Detailed Work Completed / Technical Implementation:

- Multi-Format Document Ingestion: Developed an asynchronous document processing pipeline utilizing pdf-parse, mammoth, and text sanitizers to extract clean unstructured text from PDF, DOCX, and TXT resumes.
- Google Gemini LLM Prompt Engineering: Engineered few-shot structured JSON extraction prompts using @google/genai SDK to accurately extract candidate contact info, employment timelines, educational credentials, core technical proficiencies, soft skills, and quantifiable achievements.
- Calibrated ATS Scoring Algorithm: Developed multi-factor scoring algorithms computing cosine semantic similarity between resume embeddings and target Job Descriptions (JD), factoring in required skill overlap, domain seniority, and keyword density without keyword stuffing bias.
- Automated Skill-Gap Analysis Engine: Created dynamic skill diagnostics that categorize candidate skills into Strong Matches, Partial Matches, and Critical Missing Skills, generating personalized recommendations for candidate interview readiness.

Key Deliverables & Verified Output: Asynchronous Resume Parser processing multi-page documents in under 800ms, calibrated ATS qualification matching algorithm, and visual Skill-Gap Matrix with actionable recommendations.

Guide's Remark: Excellent technical depth. The architecture diagrams, code implementations, and security standards demonstrate thorough milestone completion.		
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